

MODULE - IV

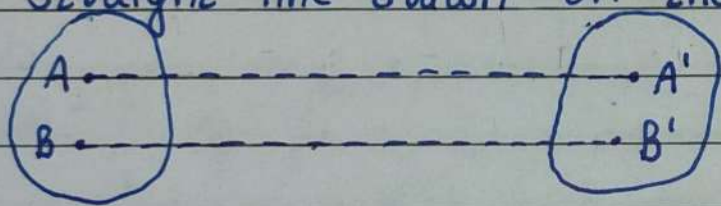
Kinematics & Kinetics

Kinematics

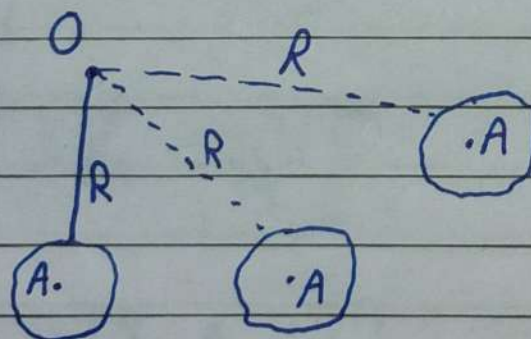
It is the branch of dynamics which deals with the motion of the bodies without referring to the forces causing the motion.

Types of Motion

1. Translation Motion: A motion is said to be translational if a straight line drawn on the moving body remains parallel to its original position at any time.



2. Rotational Motion: A motion is said to be rotational if all the particles of a rigid body move in concentric circles.



In Translational Motion, the kinematic equations are:-

$$v = u + at$$

$u \rightarrow$ Initial Velocity

$v \rightarrow$ Final Velocity

$$s = ut + \frac{1}{2} at^2$$

$s \rightarrow$ Displacement

$a \rightarrow$ Acceleration

$t \rightarrow$ Time

$$v^2 = u^2 + 2as$$

- Q. The velocity of a particle moving in a straight line is given by the expression, $v = t^3 - t^2 - 2t + 2$. The particle is found to be at a distance of 4 m from station A, After 2 sec.

Determine acceleration and displacement after 4 sec?

Solⁿ:

Given, $v = t^3 - t^2 - 2t + 2$

Now, $a = \frac{dv}{dt} = \frac{d}{dt} (t^3 - t^2 - 2t + 2) = 3t^2 - 2t - 2$

At $t = 4$ sec,

$$a = 3 \times 4^2 - 2 \times 4 - 2 = 48 - 8 - 2 = 38 \text{ m/s}^2$$

Now,

$$v = \frac{ds}{dt}$$

$$\Rightarrow v dt = ds$$

$$\Rightarrow \int v dt = \int ds$$

$$\Rightarrow \int (t^3 - t^2 - 2t + 2) dt = \int ds$$

$$\Rightarrow \frac{t^4}{4} - \frac{t^3}{3} - t^2 + 2t = s + C$$

$$\Rightarrow s = \frac{2^4}{4} - \frac{2^3}{3} - 2^2 + 2 \times 2 + C$$

$$\Rightarrow s = 1.33 + C$$

$$\text{At } s = 4 \text{ m,}$$

$$C = 4 - 1.33 = 2.66$$

$$\text{So, } s = \frac{t^4}{4} - \frac{t^3}{3} - t^2 + 2t + 2.66 = 0$$

$$\Rightarrow s = \frac{4^4}{4} - \frac{4^3}{3} - 4^2 + 2 \times 4 + 2.66 = 37.326 \text{ m.}$$

Kinetics of Rigid Body

Kinetics:

Kinetics is the branch of dynamics which deals with a motion of bodies referring to the forces causing the motion.

D'Alembert Principle (Dynamic Equilibrium)

This principle states that, the system of forces acting on a body in motion is in dynamic equilibrium with the inertia force of the body.

$$F = ma \quad \& \quad F_i = -ma$$

$$F + F_i = 0.$$